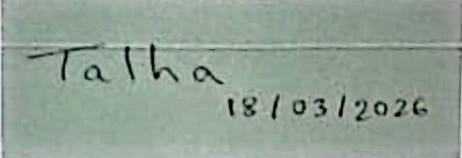
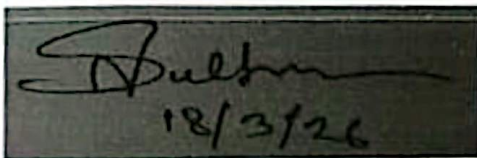


University of Dhaka
Department of Computer Science and Engineering
Professional Masters in Information and Cyber Security (PMICS)

PROJECT PROPOSAL FORM for CSE-811: Project on Cyber Security

1.	Proposed title of the project: Design and Implementation of an Automated IT Asset Management and Compliance Audit Framework for Enterprise Environments	
2.	Name of student 1: Talha Bin Alam Registration number: H-37 Contact number: 01999331739 , 01893312084	Batch: 3rd
	Name of student 2: Niger Sultana Registration number: H-49 Contact number: 01778833008 , 01766689502	Batch: 3rd
3.	Brief description of the proposed project (Importance of the project/research domain, motivation, goals, and expected outcomes – Not more than 200 words) The rapid growth of enterprise IT infrastructure has significantly increased the complexity of managing hardware, software, and network assets. Organizations often face challenges in maintaining accurate asset inventories and ensuring that systems comply with security and regulatory standards. Lack of proper asset visibility and compliance monitoring can lead to security vulnerabilities, inefficient resource utilization, and increased risk of cyber threats. This project aims to design and implement an Automated IT Asset Management and Compliance Audit Framework for enterprise environments. The proposed system will automatically discover and inventory IT assets across a network, collect hardware and software information, and perform compliance and security audits based on predefined policies. By integrating automated scanning, centralized asset databases, and reporting mechanisms, the framework will help organizations maintain accurate asset records and detect potential security risks. The motivation behind this research is to improve operational efficiency, enhance security posture, and support better decision-making in IT management. The expected outcome of the project is a functional prototype framework capable of automated asset discovery, compliance monitoring, and generation of audit reports through a centralized dashboard, assisting administrators in managing enterprise IT environments more effectively.	

4.	Methodology (Tools, methods, platforms, and techniques that will be used to carry out the project/research works – Not more than 150 words) The proposed project will follow a systematic methodology to develop an automated IT asset management and compliance audit framework enhanced with Artificial Intelligence. Initially, network asset discovery will be conducted using network scanning tools such as Nmap to identify active devices within the enterprise environment. Asset inventory tools will collect hardware and software information from the discovered systems, which will then be stored in a centralized database such as MySQL or PostgreSQL. For compliance auditing and vulnerability assessment, security scanning tools will be used to detect outdated software, missing patches, and potential security risks. Artificial Intelligence techniques will be applied using Python-based machine learning libraries such as Scikit-learn to analyze collected asset data, classify devices, and detect anomalies within the network. Automation scripts will schedule periodic scans and data processing tasks. Finally, a web-based dashboard using tools such as Grafana or a Python web framework will be developed to visualize asset information, compliance status, and AI-generated insights.	
5.	You may mention at most two names for the expected supervisor of the project	
	i) Mr. Md Bahauddin Palash,	ii) Mr. Md. Faisal Hossain
6.	You may mention at most two names for the expected co-supervisor of the project	
	i) Prof. Dr. Mamun Or Rashid	ii) Md Samiul Islam
	N.B.: A project will be guided by a supervisor and a co-supervisor, if necessary. Please note that at least one of them will be from the University of Dhaka, as determined by the concerned examination committee of the semester.	
7.	 Signature of student 1 with date	 Signature of student 2 with date

Cont
Unit
I.M.
All
1

8.

(To be filled by the chair of the concerned examination committee)

Name of the assigned supervisor:

Name of the assigned co-supervisor:

(Signature of Chair of Examination
Committee with date)